
HPL *ink* Picon Generator (Enigma2)

User manual

A guide to the tool for designing and mass-generating picons - channel icons for Enigma2 receivers.

HPL ink Picon Generator

described release: v1.35

Release date

14.07.2026

Platform

Windows 10 / 11 (desktop application)

Labelling rule: the program has two tabs - *Generator* and *Picon Sort*. Descriptions of the Generator tab refer to its four interface columns as **column 1** ... **column 4** (left to right). The **NEW in v1.35** label marks features added in the described release.

Table of contents

- | | |
|---|---|
| 1. About the program | 8. Batch - mass generation - column 3 |
| 2. Startup and access protection | 9. Send to receiver (FTP) |
| 3. Main window | 10. Picon Sort tab |
| 4. Presets | 11. Missing-picon report and the Generator bridge |
| 5. Background and composition - column 1 | 12. Undo and live preview |
| 6. Text 1 / 2 / 3 and effects - column 2 | 13. Example workflow |
| 7. Preview, saving, border, watermark and logo - column 4 | 14. Interface sections and program files |
| | 15. Troubleshooting |

1. About the program

HPL ink Picon Generator (Enigma2) v1.35 [14.07.2026] is a Windows tool that combines single-picon design and mass production for a whole channel list in one window:

- live picon design: background (colours, gradients, image file), up to three independent texts, border, decorations, logo and watermark,
- more than **130 text effects** in 8 categories - shadows, 3D, neons and glows, outlines, gradients, patterns and special effects,
- **25 background / gradient types**, including neon stages, Aurora, Bokeh, a retro-wave grid and carbon fibre,
- saving complete settings as **presets** with a thumbnail, plus export and import to a .json file,
- **batch**: load an Enigma2 channel list (.txt) and automatically generate a PNG picon for every channel - file names match the #SERVICE reference,
- uploading finished picons directly **to the receiver over FTP**,
- the **Picon Sort** tab - matching existing PNG files against a channel list, a missing-picon report and handing the missing entries over to the generator,
- a change history with an **Undo** button and the **Ctrl+Z** shortcut (up to 30 steps).

The interface uses a dark theme with an orange accent; results are generated as **PNG** files (the preview can also be saved as JPG).

Requirements

- Windows 10 or 11.
- When run from the `.py` file: Python 3 with the **Pillow (PIL)** library. The EXE build ships with all dependencies.
- The **NumPy** library is optional - when present, backgrounds and gradients render noticeably faster; without it the program automatically falls back to a slower compatibility mode.
- Local network access and the receiver's FTP account (port 21) if picons are to be uploaded to the box.

- Windows system fonts - the program builds its own catalogue of detected fonts (registry + font files).

2. Startup and access protection

First start

1. Run the program (the `hpl_picon_generator.py` file or the supplied executable).
2. The window opens centred, sized to fit the screen (up to 1860x760), with the **Generator** tab active.
3. The program immediately renders a first preview with the default settings (the caption "HPL" + "HD").

MAC address protection

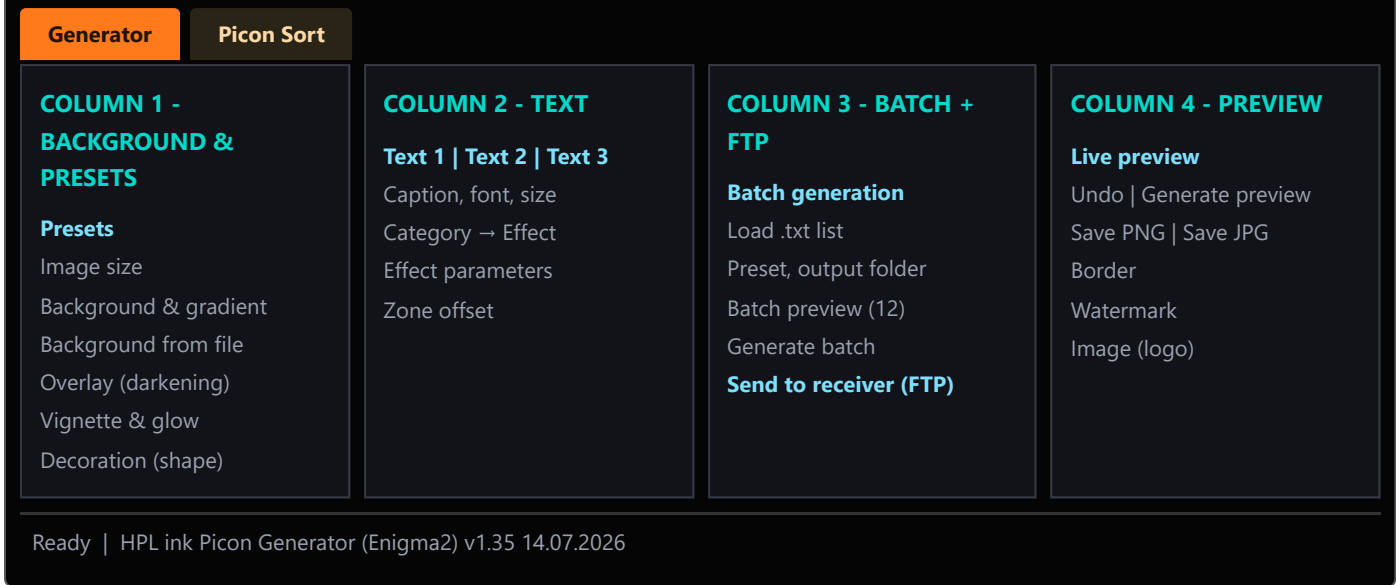
The program can be bound to specific computers. The distribution may be configured with a list of at most **5 MAC addresses**. On startup the program reads the computer's network adapter addresses using three independent methods (psutil, getmac and the system identifier), so detection also works in a compiled EXE.

MAC list configuration	Startup condition
List empty	Protection disabled - the program starts on any computer.
List filled	At least one of the computer's MAC addresses must be on the list.

Not authorised: if the computer is not on the list, the program shows "Ten adres MAC nie jest autoryzowany" (this MAC address is not authorised) and exits. In that case contact the program supplier and provide the computer's MAC address.

3. Main window

The following drawing is a privacy-safe UI diagram. It contains no IP address, password or other user data.



- The **header** shows a coloured banner with the program name plus the version number and release date.
- The **tabs** switch between designing and generating (*Generator*) and organising existing files (*Picon Sort*).
- **Columns 1-3** scroll vertically - the mouse wheel scrolls the column under the cursor; over list boxes (fonts, presets) the wheel scrolls the list itself.
- The **status bar** at the bottom shows the result of the last operation, batch progress and FTP messages.

Live preview: almost every settings change (slider, field, colour, effect) automatically refreshes the preview after a short idle moment. There is no need to click "Generate preview" every time.

4. Presets

The **Presets** section at the top of column 1 stores and restores the *complete* set of settings: size, background (including the background image path), texts 1-3 with effects, border, decoration, vignette, watermark and logo.

Working with presets

- **Preset list** - selecting a preset shows a generated thumbnail of it below the list. **NEW in v1.35**
- **Load** - restores the selected preset's settings (effect categories and parameter section visibility adjust automatically).
- **Save preset** - stores the current settings under the name from the **Name** field. Saving under an existing name overwrites that preset.
- **Delete** - removes the selected preset.
- **Random style** - randomises the background, gradient, colours and the Text 1 effect; a convenient starting point for quick prototyping.
- **Export / Import** - saves all presets to a chosen `.json` file and loads presets from a file (for duplicates the program asks before overwriting).

Built-in presets

Preset	Description
CANAL+ 360 neon	A neon stage with a gradient letter outline - the CANAL+ 360 picon style. Also works in batch (the channel name replaces the "360" caption).
Codex Sol NEW in v1.35	A composition on the "Neon studio - wet floor" background with the "Two-zone neon + reflection" effect, a neon bar at the top and a third text "CANAL+".

Safe saving: the preset file `hpl_picon_presets.json` is written atomically (a temporary file first, then a swap). If the preset file turns out to be corrupted, the program creates a `.bak` copy instead of silently overwriting the data. The "saved" message appears only after the file has actually been written.

5. Background and composition - column 1

Image size

There are 11 sizes to choose from: 100x60, 128x128, 220x132, 256x256, 320x180, 400x240 (default), 512x512, 600x360, 800x450, 1280x720 and 1920x1080 px.

Background and gradient

The background is built from up to three colours (**Background colour #1 / #2 / #3**) and a **Gradient type** - 25 variants:

- classic: solid background, horizontal, vertical and diagonal (\ and /) gradients, radial (centre / corner / ellipse), 3-colour multi-stop, diamond, conical, wave,
- textural: horizontal and vertical stripes, rays, checkerboard, marble, plasma, noise, spiral / vortex,
- scenic: **Neon stage (3 colours)**, **Neon studio - wet floor (3 colours)** with fog, a horizon and reflections
NEW in v1.35, **Aurora (3 colours)**, **Bokeh**, **Retro-wave grid**, **Carbon (carbon fibre)**.

Background from file

1. Tick **Enable background from file** and click **Load image** (PNG/JPG).
2. Adjust **Background opacity** (0-100%), **Background zoom** (0.1-5.0x) and **Offset X / Y** (±2000 px).
3. **Gradient opacity on background** controls how strongly the colour gradient covers the image.
4. The most convenient way to position the image is directly in the preview: **dragging with the mouse moves the background, the mouse wheel changes the zoom**.
5. **Remove background** detaches the image from the project.

Overlay, vignette and glow

- **Overlay (darkening)** - colour + opacity 0-100%; a uniform layer over the whole background, e.g. to dim a photo under the text.
- **Darken corners** - a vignette that works on every background, including an image file.
- **Brighten centre** - a subtle glow in the middle of the picon.

Decoration (shape)

A decorative shape drawn on the background: *Diagonal bar*, *Horizontal bar (bottom)*, *Vertical bar (left)*, *Circle (corner)*, *Arc (bottom)*, *Accent line (corner)*, *Triangle (corner)* and the *Neon bar (top)* with a two-colour outline and glow **NEW in v1.35**. Each shape has a colour, opacity, size and a **Behind text** option.

6. Text 1 / 2 / 3 and effects - column 2

Three independent texts

Tab	Purpose
Text 1 (orange)	The picon's main caption (default "HPL"). Replaced with the channel name in batch.
Text 2 (green)	A suffix, "HD" by default. In batch it is set to "HD" whenever the channel name ends with HD.
Text 3 (blue)	A free position - centred on the whole image and moved with the Offset X/Y fields; empty by default. Ideal for a fixed caption such as a platform name.

The **Copy to Text 2** button copies the Text 1 settings (without the caption, size and font colour) - the fastest way to give both texts a consistent style.

Caption and font

- **Caption** - a multi-line field; Enter starts another caption line.
- **Family** - a list of fonts detected in the system with a filter box (the program shows the number of detected fonts). Bold/Italic use real font variants and fall back to synthetic slanting and thickening when a variant is missing.
- **Size** 6-300 px, **Bold**, **Italic**, **Text colour**, **Opacity** 0-100%, **Text rotation** $\pm 180^\circ$.

Text effect - two-level selection

Instead of a single list of over 130 entries, the effect is chosen in two steps: **Category** → **Effect**.

Category	Example effects
Shadows	Classic, soft, perspective, long shadow, gradient shadow, bottom mirror shadow, floor-cast shadow.
3D	Layered 3D, 3D with side gradient, extruded 3D, chromed 3D.
Glow	Glow, double neon, neon tube, cyberpunk neon, UV / blacklight, electric arc, rainbow glow, cracked neon, Two-zone neon and Two-zone neon + reflection NEW in v1.35 .
Outlines	Outline, multiple outline, neon outline, contour (hollow centre), gradient neon contour, two-colour double contour.

Category	Example effects
Gradients on text	More than 40 palettes: rainbow, ocean, matrix, aurora, lava, gold, chrome, copper, titanium, oil-slick chrome and more.
Patterns on text	Stripes, scanlines, pixel / 8-bit, grid, dots, hex, halftone, pop art, bricks, marble, camouflage.
Special	Fire, ice, hologram, RGB glitch, comic, knockout (caption cut from the background), neon badge, sticker, scorch, 3D glass (glassmorphism), duotone, flickering neon.

Contextual parameters

Below the effect list, only the parameter sections actually used by the selected effect are shown:

- **Shadow** - colour, distance 1-40 px, angle 0-359°, blur 0-30 px,
- **Outline** - colour and width 1-12 px,
- **Glow / Neon** - colour and radius 2-40 px,
- **3D** - side colour and depth 1-20 px,
- **Effect colours** - Colour #1 / #2 / #3 for gradients and patterns,
- **Gradient direction** - 10 directions (vertical, horizontal, diagonals, radials, diamond, conical, wave, plasma) plus the **Gradient shift** slider 0-100%, which cyclically shifts the palette phase and creates new variants of every gradient.

The **Zone offset** section (X/Y, ±500 px) moves the text relative to its default position.

7. Preview, saving, border, watermark and logo - column 4

Preview

- Dragging with the **left** mouse button moves the background image; the **wheel** changes the background zoom.
- Dragging with the **right** mouse button moves the logo.

Buttons

- **Undo** - restores the state before the last change (details in chapter 12).
- **Generate preview** - forces an immediate redraw.
- **Save PNG / Save JPG** - saves the current preview to a file (JPG at quality 95). If no preview has been generated yet, the program asks for one first.

Border

After ticking **Enable border**, 8 styles are available: *Full, Rounded, Double, Dashed, Corners only, Neon (glow), Bevelled 3D, Gradient*. The shared parameters are colour, width 1-20 px and edge inset 0-30 px. Contextual parameters appear only for styles that use them: corner radius 0-40 px, and for the Gradient style additionally Colour #2 and the border gradient direction (horizontal, vertical, diagonal, perimeter).

Watermark

The "*ink*" mark (red italic) in a corner of the picon. It can be **switched on and off**, and the setting is stored in the preset **NEW in v1.35**. The **watermark corner** can be bottom-right or bottom-left.

Image (logo)

1. Tick **Enable** and click **Load image** (PNG/JPG with transparency).
2. Set the **logo scale** 5-100%, **position X / Y**, **logo opacity** and the **Behind text** option.
3. The logo can also be moved by dragging with the **right mouse button** in the preview.

Paths in presets: a preset remembers the background and logo file paths. Batch will use those files as long as they still exist on disk.

8. Batch - mass generation - column 3

Channel list format

Batch accepts `.txt` files in the Enigma2 list format - pairs of `#SERVICE` and `#DESCRIPTION` lines. The parser handles:

- a channel name embedded in the `#SERVICE` line itself (ending `::name`),
- `#SERVICE` lines without a name (a single trailing colon - typical Enigma2 bouquets; the name is taken from `#DESCRIPTION`),
- `#DESCRIPTION` lines appearing both *after* and *before* the `#SERVICE` line.

For each channel the program creates a `PNG` file whose name equals the `#SERVICE` reference with colons replaced by underscores, e.g. `1_0_1_3DEA_640_13E_820000_0_0_0.png`.

Automatic texts: the channel name is written in CAPITALS as Text 1; if it ends with "HD", the suffix goes to Text 2. The program automatically fits the Text 1 font size to the text zone and, when needed, wraps long names onto at most two lines.

Step by step

1. **Load .txt list** - the label next to the button shows the number of channels read.
2. Choose the **preset to use** from the drop-down list (saved + built-in presets).
3. **Choose the output folder** for the PNG files.
4. Optionally tick **Skip existing files** - useful when resuming an interrupted run.
5. **Batch preview (first 12)** - a window with a thumbnail grid of the first 12 channels rendered with the selected preset; lets you judge the style before the full run.
6. **Generate batch** - generation runs in the background; the progress bar shows the percentage, a file counter and the **estimated time of completion (ETA)** computed from the files actually generated.

Errors on individual channels do not stop the batch - after finishing, the program writes a `_batch_errors.txt` log in the output folder and shows the error count on the status bar.

9. Send to receiver (FTP)

The section at the bottom of column 3 uploads the generated picons directly to an Enigma2 receiver.

1. Enter the receiver's **Host / IP**, **Login** (default `root`) and **Password** (masked; VU+ receivers always require a password).
2. Choose or type the **picon path on the receiver**. The list suggests three typical locations:
`/usr/share/enigma2/picon`, `/media/hdd/picon`, `/media/usb/picon`.

3. **Test connection** - logs in to FTP (port 21), enters the given folder and shows the number of PNG files it contains. A successful test saves the configuration.
4. **Send to receiver** - uploads all PNG files from the batch output folder. Progress ("Sending X / Y...") is shown next to the FTP section and the result also appears on the status bar.

Password safety: the FTP configuration is stored in `hpl_picon_ftp.json` and the password is protected with **Windows DPAPI** - only the same user on the same computer can read it. An older (base64) configuration is migrated automatically on first use.

Upload requirements: the batch output folder with PNG files must exist before uploading. After uploading the picons, reload the channel list or restart the Enigma2 GUI so the receiver picks up the new icons.

10. Picon Sort tab

Picon Sort organises existing picon collections: it matches PNG files against a channel list and points out what is missing.

1. Choose the **folder with PNG icons** (e.g. an unpacked picon pack).
2. Choose the **.txt list file** (the same syntax as in batch - chapter 8).
3. Click **Match and copy**.

For every channel the program looks for a `<ID_with_underscores>.png` file and copies the found ones into the **Znalezione** ("Found") subfolder in the program directory. The results go to a table:

Column	Content
Status	OK - file found and copied; MISSING - no file for the channel.
Channel name	The name from the list (#DESCRIPTION or #SERVICE).
File	The expected PNG file name.
Info	Operation details (copied / not found).

A summary is shown above the table: *Found / Missing / Copied*.

11. Missing-picon report and the Generator bridge

The **Missing report (N)** button opens a window on the right half of the screen listing the entries without a PNG file: position number, channel name, the underscore identifier and the expected file name.

- **Copy identifier** - copies the selected entry's ID to the clipboard (e.g. to search for a picon online manually).
- **Generate missing picons** - the report's key feature: it moves all missing entries straight into the Generator's batch list (with the CAPITALS + "HD" → Text 2 rules) and switches the program to the Generator tab. Just pick a preset and folder, then click **Generate batch** to produce only the missing picons.

12. Undo and live preview

- The history covers **up to 30 steps** and remembers the full settings state, including the loaded background and logo files.
- A state is recorded automatically after every successful preview render; a burst of rapid changes (e.g. dragging a slider) is merged into a single history step.
- Undo is available via the **Undo** button next to "Generate preview" or the `Ctrl+Z` shortcut.
- The `Ctrl+Z` shortcut is inactive while typing in text fields - it does not conflict with the system text undo.

- When the history is empty, the Undo button stays greyed out.

13. Example workflow

A typical scenario: a complete picon set for your own channel list in a single style.

1. Style

Design a picon in the Generator tab or load a preset (e.g. "Codex Sol") and fine-tune the colours.

2. Preset

Save the settings with "Save preset" under your own name.

3. Batch

Load the .txt list, pick the preset and folder, check "Batch preview", run "Generate batch".

4. Receiver

In the FTP section test the connection and send the picons to the receiver.

The tidy-up variant: first **Picon Sort** matches an existing picon pack against the list, then the missing report's **Generate missing picons** button produces only the missing icons in the Generator - in the same style as the rest.

14. Interface sections and program files

Section cheat sheet

Section	Location
Presets, Image size, Background & gradient, Background from file, Overlay, Vignette & glow, Decoration	Generator - column 1
Text 1 / Text 2 / Text 3, Font, Text effect, parameters, Zone offset	Generator - column 2
Batch generation, Send to receiver (FTP)	Generator - column 3
Preview, Undo / Generate preview / Save PNG / JPG, Border, Watermark, Image (logo)	Generator - column 4
Match and copy, Missing report, results table	Picon Sort tab

Program files

All files are created in the program directory (next to the `.py` / `.exe` file):

File / folder	Purpose
<code>hpl_picon_presets.json</code>	Saved presets. If the file gets corrupted, a <code>.bak</code> copy is created.

File / folder	Purpose
hpl_picon_ftp.json	FTP configuration; the password is protected with Windows DPAPI.
Znalezione	The subfolder with picons copied by Picon Sort.
_batch_errors.txt	The error log of the last batch - in the batch output folder.
icon.ico	The program window icon (optional).

15. Troubleshooting

"This MAC address is not authorised"

The computer is not on the list of authorised MAC addresses. Contact the program supplier and provide the MAC address (command line: `getmac`).

The preview renders slowly

Install the NumPy library (`pip install numpy`) - backgrounds and gradients then render many times faster. Large sizes (1280x720, 1920x1080) and blur-based effects always take longer; use a smaller size for experiments and set the target size before the batch run.

The "Generate batch" button does not start

Check three conditions: a loaded channel list (the label shows the channel count), a preset selected in the drop-down and a chosen output folder. The program shows a message pointing at the missing element.

Batch reports errors for individual files

Open `_batch_errors.txt` in the output folder - each line contains the channel name, the identifier and the cause. The most common causes are an unavailable background/logo path stored in the preset or a font missing from the system.

FTP: "login/path error" or "connection error"

1. Check the receiver's IP and that the receiver is switched on (not in deep standby).
2. The login is usually `root`; a password is required - it is set in the receiver's menu (e.g. OpenWebif / network settings).
3. Verify that the picon path exists on the receiver; create the `picon` folder if needed.
4. The computer's firewall must not block outgoing connections on port 21.

Picon Sort finds nothing

The file names in the folder must be #SERVICE identifiers with underscores (e.g.

`1_0_1_3DEA_640_13E_820000_0_0_0.png`). If the picon pack uses channel names instead of identifiers, ID matching is not possible - generate the picons with batch instead.

Text 2 does not appear in batch

Text 2 is filled only when the channel name ends with the word "HD". Any other Text 2 content from the preset is overwritten during batch.

Invalid colour in a field

Colours use the `#RRGGBB` format. An invalid or empty value is replaced with white instead of raising an error.

Presets disappeared after a crash

Check whether a `hpl_picon_presets.json.bak` copy exists next to `hpl_picon_presets.json` - it holds the last version from before the corruption; just remove the `.bak` extension.